

**CARIBBEAN EXAMINATIONS COUNCIL
SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 01 – General and Technical Proficiencies

THEORY

1½ hours

29 MAY 2008 (p.m.)

INSTRUCTIONS TO CANDIDATES

1. This paper has four sections.

SECTIONS ONE AND TWO ARE TO BE DONE BY ALL CANDIDATES.

TECHNICAL PROFICIENCY CANDIDATES MUST DO SECTIONS ONE, TWO, AND THREE.

GENERAL PROFICIENCY CANDIDATES MUST DO SECTIONS ONE, TWO, AND FOUR.

2. Number EACH answer correctly in the answer booklet.
3. You may write Basic or Pascal (or any other language) code in those questions requiring pseudocode/algorithm.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTIONS I and II

TECHNICAL AND GENERAL PROFICIENCIES

ANSWER ALL QUESTIONS.

SECTION I – 30 marks

1. Use the figures A to D to answer the questions below.

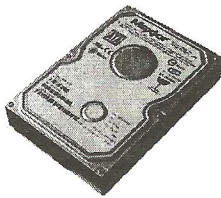


Figure A

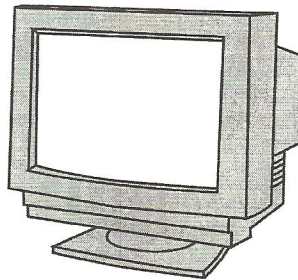


Figure B

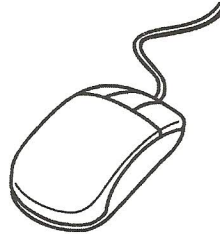


Figure C

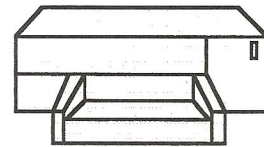


Figure D

- (a) Write the terms Figure A to D on your paper. Then classify EACH figure, as an input, output or storage device. (4 marks)
- (b) Give the names of the devices labelled
- (i) Figure A
 - (ii) Figure B
 - (iii) Figure C
 - (iv) Figure D
- (2 marks)

Total 6 marks

2. (a) Explain how a general-purpose application package is different from a specialized application package. (2 marks)
- (b) Write the numbers 1 to 4 in your answer booklet. Match the number for the application in Table 1 with the letter for the type of software in Table 2 that BEST describes that application.

Table 1

	Application
1	AutoCad for Draftsmen
2	LittleRed Pizza Delivery System
3	MS Word
4	Linux

Table 2

	Types
A	Custom-written software
B	Operating system software
C	Specialized software package
D	General purpose software

(4 marks)

Total 6 marks

3. You are the technician in an office complex. You are called because an employee who has a colour inkjet printer has tried repeatedly to print a document that has both text (in black) and a chart (in colour). The chart prints in colour, but the text prints faded in some places and appears in streaks in other parts of the document.
- (a) Give TWO possible reasons why this might have happened. (2 marks)
- (b) After this problem is fixed the employee calls again to say that the printer does not print on a multi-part form.
- (i) Give the reason for the printer not printing on the second sheet of the form. (1 mark)
- (ii) What kind of printer would have been more appropriate for this task? (1 mark)
- (c) You ask the employee to provide you with a hardcopy and a softcopy of the document she is trying to print. She is not sure what you mean. Explain the terms 'hardcopy' and 'softcopy'. (2 marks)

Total 6 marks

GO ON TO THE NEXT PAGE

4. The following is a directory listing using MSDOS.

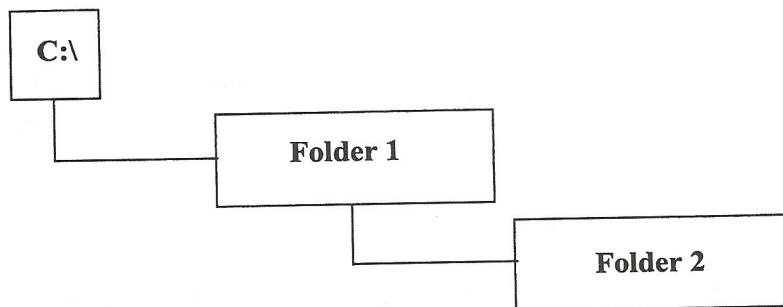
```
C:\ExamFile\Questions>dir
Volume in drive C has no label.
Volume Serial Number is 0000-33D7

Directory of C:\ExamFile\Questions

11/23/2006  02:33 AM    <DIR>          .
11/23/2006  02:33 AM    <DIR>          ..
11/21/2006  08:45 AM                27,648 Paper1.doc
11/22/2006  09:37 PM                30,720 Paper2.doc
               2 File(s)              58,368 bytes
               2 Dir(s)              177,446,912 bytes free

C:\ExamFile\Questions>
```

- (a) State the type of user interface that is illustrated in the MSDOS printout above. (1 mark)
- (b) The diagram below illustrates the directories or folders from the above listing. State the names of Folder 1 and Folder 2.



- (c) Name the directory or folder whose contents are being listed. (1 mark)
- (d) What is the TOTAL size of files in the "*Questions*" directory? (1 mark)
- (e) How many files would you have in the Questions directory if you copied both Paper1.doc and Paper2.doc files to a new subdirectory "*Section1*"? (1 mark)
- (f) How many files would you have in the Questions directory if you moved both Paper1.doc and Paper2.doc files to a new subdirectory "*Section1*"? (1 mark)

Total 7 marks

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5. Every month, a utility company takes readings from the meters of homeowners. The company then processes all readings to calculate every homeowner's monthly bill, prints the bills and posts them.
- (a) State the name given to the type of processing that allows multiple tasks to be processed as a group. (1 mark)
 - (b) Give ONE benefit of this type of processing. (2 marks)
 - (c) Give ONE problem with this type of processing. (1 mark)
 - (d) State whether this type of processing by the company is similar to simplex, half-duplex or full-duplex transmission. (1 mark)

Total 5 marks

SECTION II – 30 marks

6. Computer-related crimes are on the rise. One form of computer-related crime is electronic eavesdropping.
- (a) Explain the term 'electronic eavesdropping'. (2 marks)
 - (b) You go to great lengths to secure a word document that you want to share *only* with persons in your department. This document can be edited by persons in your department.
 - (i) Identify ONE physical access restriction that would help to ensure that **ONLY** authorized members in the department have access to important computer equipment. (1 mark)
 - (ii) Identify ONE security measure that would ensure that **ONLY** authorized members in the department have access to the word document. (1 mark)
 - (iii) Explain why a sharing restriction can be useful when one person is accessing the word document. (2 marks)

Total 6 marks

GO ON TO THE NEXT PAGE

7. Consider the use of teleconferencing and telemarketing in a business.
- (a) State ONE similarity between the two. (1 mark)
 - (b) State ONE difference between the two. (1 mark)
 - (c) Explain how telecommuting is different from both teleconferencing and telemarketing. (1 mark)

Table 3 shows a list of acronyms labelled P to R and Table 4 shows a list of functions for the acronyms labelled 1 to 3.

Table 3

Acronym	
P	CADD
Q	CAE
R	CAM

Table 4

Functions	
1.	Assists in data analysis
2.	Aids in the precision of product manufacturing
3.	Creates three-dimensional drawings

- (d) Write the letters P, Q and R in your answer booklet. Match the letter with the number of the correct definition from Table 4. (3 marks)

Total 6 marks

8. I use my credit card to order a flat screen TV from *www.mydeals.com*.
- Within a day, an email arrives asking that I verify the order.
 - A day later, I am sent another verification email which asks that I submit the last four digits of the credit card I used in my order.
 - Five days later I receive the product I ordered.
- (a) Give ONE reason why the company would ask me to verify the order within the first day. (1 mark)
 - (b) Give ONE reason why the company would send an email asking me to re-submit the last four digits of my credit card. (1 mark)
 - (c) Give TWO reasons why this action qualifies as an e-commerce activity. (2 marks)
 - (d) Three departments are involved in the successful completion of the order process. If the Dispatch Department is responsible for product delivery, then state the departments that are responsible for requesting credit card verification and verifying the order. (2 marks)

Total 6 marks

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9. Consider the following URL:

http://www.youthalive.org/index.html

- (a) What part of this address would be considered the website? (1 mark)
- (b) What part of the address would be the webpage? (1 mark)
- (c) What does the abbreviation 'html' represent? (1 mark)
- (d) What is meant by 'http'? (1 mark)
- (e) What is a 'URL'? (1 mark)
- (f) Create a URL for a company called Rising Sun. (1 mark)

Total 6 marks

10. A consultant has encouraged the administration of the bank where you work to install a MICR system.

- (a) What does the term 'MICR' mean? (1 mark)
- (b) State ONE function of an MICR system that would be useful to a bank. (1 mark)
- (c) Before agreeing to install the MICR system, the Systems Analyst and the Network Manager must determine:
 - the benefits this new feature will bring to the bank and
 - the locations of servers and other devices to support the MICR

Explain which of the above tasks would be assigned to

- (i) the Systems Analyst (2 marks)
- (ii) the Network Manager. (2 marks)

Total 6 marks

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TECHNICAL PROFICIENCY

THIS SECTION IS FOR TECHNICAL PROFICIENCY CANDIDATES ONLY.

SECTION III – 30 marks

PROGRAMMING

ANSWER ALL QUESTIONS.

11. Copy the labels EX1, EX2, and EX3 on separate lines in your answer booklet.
- (a) Write the corresponding class of programming language, PL1 to PL4, which matches the appropriate example, EX1 to EX3.
- EX1: MUL 12, 10
EX2: 00100101
EX3: READ name
- PL1: Fourth generation
PL2: High level
PL3: Machine level
PL4: Assembly Level (3 marks)
- (b) On the same line of the examples EX1, EX2, and EX3 in your answer booklet, identify the generation of programming language used. (3 marks)

Total 6 marks

12. For EACH of the following tasks, state whether it is described as input, output, processing or storage:
- (a) (i) Calculating the cost of a pair of shoes after 10% discount ✓
(ii) Dialing the number of the store to ask if a shoe size is in stock
(iii)✓ Reading about a shoe sale in the newspaper ✓
(iv)✓ Saving the money to buy the pair of shoes ✓
(v) Asking the salesperson the discount on a pair of shoes (5 marks)
- (b) Place the tasks labelled (i) to (v) in Part (a) above in the CORRECT sequence. (3 marks)

Total 8 marks

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13. (a) State what EACH of the following operators means:

- (i) $\geq$
- (ii) $=$
- (iii) $<$
- (iv) $\leq$

(4 marks)

(b) State the output of the following algorithms, if MARK is 50:

- (i) READ MARK
IF mark $<$ 40
THEN PRINT 'The student has failed'
ELSE PRINT 'Let us see'
- (ii) WHILE (mark $\leq$ 50) DO
BEGIN
MARK = MARK + 2
PRINT 'Working'
END WHILE
PRINT 'The student has passed'

(1 mark)

(1 mark)

Total 6 marks

14. Use the fragment of code below, to answer Parts (a) to (c) that follow:

Line 1 A = 3
Line 2 B = 5
Line 3 Sum = 1
Line 4 While B $\leq$ 10
Line 5
Line 6 B = B + A
Line 7 Sum = Sum + B
Line 8 End while
Line 9 PRINT Sum

(a) For EACH of the following, identify ONE line which contains:

- (i) An assignment statement
- (ii) A loop
- (iii) An output statement

(3 marks)

(b) Copy and complete the trace table below:

A	B	Sum
3	5	1

(6 marks)

(c) State the result of the algorithm.

(1 mark)

Total 10 marks

GO ON TO THE NEXT PAGE

GENERAL PROFICIENCY

THIS SECTION IS FOR GENERAL PROFICIENCY CANDIDATES ONLY.

SECTION IV – 30 marks

PRODUCTIVITY TOOLS

ANSWER ALL QUESTIONS.

15. The questions which follow the paragraph below relate to Word Processing.

Information Technology has evolved over the past five decades in response to the need for more efficient techniques to manage the significantly increased volume and sophistication of the knowledge reservoir of mankind. It merges the study of Computer Science, Telecommunications and Office Automation; involves the collection, storage, accessing, processing and dissemination of information and impacts on both work and leisure activities.

- (a) What kind of justification is used in this paragraph? (1 mark)
- (b) Use the phrase *Technology and Me* as a title for this paragraph. What steps would you take to place it above the paragraph and centre it? (2 marks)
- (c) How can you change the paragraph so that the second sentence is in a new paragraph? (1 mark)
- (d) If you have successfully placed the second sentence into a paragraph of its own, explain how you would centre-justify that entire paragraph. (1 mark)
- (e) Describe how you would move the entire second sentence to make it the first sentence. (2 marks)
- (f) Some words are misspelled in this paragraph. What feature of word processing would you use to correct them? (1 mark)
- (g) What steps would you take to change the line spacing in your two paragraphs from single to double spacing? (2 marks)

Total 10 marks

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16. The questions (a) to (h) relate to the Spreadsheet.

	A	B	C	D	E
1	GUYEXPO				
2	EXPORTS: 2005 - 2007				
3					
4	Products	2005	2006	2007	TOTAL
5	Rice	30000	50000	40000	
6	Sugar	50000	70000	80000	
7	Bauxite	5000	7000	6000	
8	Timber	7000	8000	3000	
9	TOTAL				

- (a) Write the function that would calculate the TOTAL rice exported during the period 2005 – 2007. (2 marks)
- (b) State how you would centre the text “GUYEXPO” across the columns used by the spreadsheet. (2 marks)
- (c) State the range that would show the TOTAL exports for EACH of the three years. (1 mark)
- (d) State the cell that contains the amount of sugar exported in 2007. (1 mark)
- (e) Write the formula to compute the percentage of exports for 2005 that was contributed by rice. (1 mark)
- (f) The comma (,) format was applied to the numeric data in columns B, C, D and E. What should happen? (1 mark)
- (g) It was suggested that the headings for all numeric columns should be aligned to the right. Give ONE reason for this. (1 mark)
- (h) Formulae were placed in Column E and Row 9 to compute totals. Suppose the data in C7 was changed from 7000 to 8000, state three cells whose data would change automatically. (1 mark)

Total 10 marks

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17. The questions which follow the paragraph below relate to Database Management.

You are required to create a database table to store information on the students in a class. You are required to keep track of the following data for each student: student ID, student last name, student first name, date of birth, year of entry.

- (a) Create a structure for the above table. Your structure should include field names, data types and field size. (3 marks)
- (b) Write ONE example of a student record. (1 mark)
- (c) Identify the database feature that would arrange the records in DESCENDING order by year of entry. (1 mark)
- (d) Explain how ANOTHER field called grade point average (GPA) can be included in the database table. (2 marks)
- (e) What criteria you would use to search the table for ALL students who have been admitted after 2005? (1 mark)
- (f) Explain how you would generate a report to show ALL students ordered by last name, according to their year of entry. (2 marks)

Total 10 marks

END OF TEST



TEST CODE **01529010**

FORM TP 2009082

MAY/JUNE 2009

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SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 01 – General and Technical Proficiencies

THEORY

1½ hours

01 JUNE 2009 (p.m.)

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1. This paper has four sections.

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GENERAL PROFICIENCY CANDIDATES MUST DO SECTIONS I, II, AND IV.

2. Number EACH answer correctly in the answer booklet.
3. You may write Basic or Pascal (or any other language) code in those questions requiring pseudocode/algorithm.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01529010/F 2009



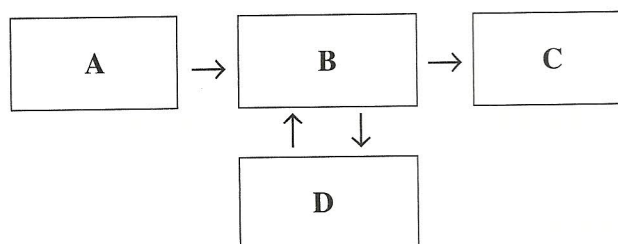
SECTIONS I and II

TECHNICAL AND GENERAL PROFICIENCIES

ANSWER ALL QUESTIONS.

SECTION I – 30 marks

1. (a) Write the labels A, B, C, and D on separate lines, one below the other, in your answer booklet. Then write the term *input*, *output*, *processing* or *storage* next to the letter that best describes the stage as shown in the diagram below.



(4 marks)

- (b) For EACH of the terms input, output, processing and storage, write ONE example of a device that is used to perform this operation. **(4 marks)**

Total 8 marks

2. Floppy disks were once the usual medium for saving files.

- (a) Describe ONE example where it would still be sensible to use a floppy disk for software distribution. **(1 mark)**
- (b) State TWO different **devices**, other than floppy disks, that are used to store files. **(2 marks)**
- (c) Using the two devices that you chose in 2 (b), state which ONE might hold more files. **(1 mark)**
- (d) Suppose you decided to e-mail your work to your personal e-mail address. Describe ONE advantage and ONE disadvantage of doing this. **(2 marks)**

Total 6 marks

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3. Compute the following:

- (a) The **eight-bit** binary representation of the decimal number 14. (1 mark)
- (b) The sign-and-magnitude representation for the decimal number **NEGATIVE** 14. (1 mark)
- (c) The result of $55 - 25$, using eight-bit two's complement representation. Express your answer as an **eight-bit** binary number. (3 marks)
- (d) The decimal equivalent for the **binary coded decimal** number 0011 0110 0000. (2 marks)

Total 7 marks

4. You have hired a company to create a software package called ABC specifically for your business.

- (a) State the **general name** given to this type of software package. (1 mark)
- (b) Suppose you bought a package similar to ABC, called FGH, 'off-the-shelf'.
 - (i) Explain what is meant by 'off-the-shelf'. (1 mark)
 - (ii) Describe ONE advantage and ONE disadvantage EACH of
 - a) hiring someone to create ABC
 - b) buying FGH. (4 marks)
- (c) Suppose you wished to use ABC to integrate with your other software packages used in the business.
 - (i) Explain why you would want to integrate them. (2 marks)
 - (ii) Give ONE disadvantage that you might have in integrating these programs. (1 mark)

Total 9 marks

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SECTION II – 30 marks

5. Write the list of applications A to E from Table 1 in your answer booklet. Match EACH application with the MOST appropriate input device from Table 2 and write the number of the device chosen next to the application.

Table 1

	APPLICATIONS
A	Point-of-sale
B	Games
C	Architectural design
D	Multiple-choice examination
E	Check processing

Table 2

	INPUT DEVICES
1	optical mark reader
2	barcode reader
3	magnetic ink character recognition
4	joystick
5	plotter

Total 5 marks

6. (a) Identify TWO industries in which robots are useful. (2 marks)
- (b) State ONE advantage EACH of using robots in EACH industry identified in 6 (a). (2 marks)
- (c) State ONE disadvantage of using robots in industry. (1 mark)

Total 5 marks

7. Consider the following encryption code where each letter of a password represents a different letter in the alphabet (A is represented as G, B as H, C as I and so on).

Password letters	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
Encryption letters	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R

- (a) (i) What is the encryption code for the password **HIKE**? (1 mark)
- (ii) If the encrypted password is represented as **RED** what is the real password? (1 mark)
- (b) What is the purpose of having a password? (1 mark)
- (c) Within many organizations, data can be stored in a central location where staff in various departments can have access to them. Describe TWO problems that may arise because of sharing these data. (2 marks)

Total 5 marks

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8. (a) James is a **computer trainer** while Rachel is a **systems analyst**. Give ONE function of EACH of these roles. (2 marks)

(b) Give the meaning of the following terms:

(i) Telemarketing (1 mark)

(ii) Telecommuting (1 mark)

(iii) Teleconferencing (1 mark)

Total 5 marks

9. (a) Explain what is meant by 'computer-related crime'. (1 mark)

(b) Give TWO examples of computer-related crime. (2 marks)

(c) Give TWO steps that can be taken to help prevent computer-related crime. (2 marks)

Total 5 marks

10. The *Computers for Caribbean Schools* project provides at least one computer for all classrooms.

(a) State TWO ways in which the computers can be used in the classroom. (2 marks)

(b) One class added two other devices to their computer and can now ^{capture} ~~store~~ copies of their own photographs, create greeting cards and print them. State the TWO devices that were added. (2 marks)

(c) One class connected their computer to the Internet. List ONE way in which they can protect their computer from getting viruses. (1 mark)

Total 5 marks

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TECHNICAL PROFICIENCY

THIS SECTION IS FOR TECHNICAL PROFICIENCY CANDIDATES ONLY.

SECTION III – 30 marks

PROGRAMMING

ANSWER ALL QUESTIONS

- 11.** (a) Draw a flowchart to represent the following algorithm:
- (i) Prompt the user to enter an integer (1 mark)
 - (ii) Read the integer and store in variable N (1 mark)
 - (iii) Divide the number by 2 and store the remainder in variable ANS (1 mark)
 - (iv) If ANS is
 - a) zero (0), multiply N by itself ($N*N$) and place the result in variable Y
 - b) not zero (0), multiply N by 2 ($N*2$) and place the result in Y (3 marks)
 - (v) Print the answer (Y). (1 mark)
- (b) Calculate and write the value that will be printed if the number input is 13. (1 mark)

Total 8 marks

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12. Consider the following algorithm:

1. Identify and initialize variables
 $a \leftarrow 2, d \leftarrow 0$
2. Begin Processing
3. **If** $a > 5$
4. $d \leftarrow 2 * (d + 1)$
5. else
6. **while** $a < 5$
7. begin
8. $d \leftarrow d + a$
9. $a \leftarrow a + 1$
10. end
11. $Y \leftarrow d * d$
12. **Print** Y
13. End Processing

Step	d	a
0	0	2
1		
2		
3		
4		

Table 3

- (a) Copy the trace table (**Table 3** above) into your answer book. Show how EACH value in the TWO variables (a, d) changes, during processing. (6 marks)
- (b) Calculate the value of Y in line 11. (2 marks)
- (c) Using the four lines highlighted in the algorithm (3, 6, 8, 12), write the line number that corresponds to the following labels:
 - (i) Output statement (1 mark)
 - (ii) Arithmetic operation (1 mark)
 - (iii) Conditional statement (1 mark)
 - (iv) Loop (1 mark)

Total 12 marks

13. Write a structured program using the following algorithm:

- (a) Prompt a user to enter an integer value (2 marks)
- (b) Store the value in a variable *num* (2 marks)
- (c) Multiply *num* by *num*, and put the result in *numsqr* (2 marks)
- (d) Print the result if it is less than or equal to 1000. Otherwise, print the following message:
 THE OUTPUT IS VERY LARGE (4 marks)

Total 10 marks

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GENERAL PROFICIENCY

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SECTION IV – 30 marks

PRODUCTIVITY TOOLS

ANSWER ALL QUESTIONS

14. The following questions relate to Word processing, and to the envelope below:

Edu-Training Limited

25, 54th Avenue Road

St. Georges

McKhan Street

NEVIS

>>>Your future training needs

- (a) State whether the word wrap feature can be seen in any text in the envelope. (1 mark)
- (b) Name TWO formatting features used in the document. (2 marks)
- (c) Write TWO types of alignments, used in the text above. (2 marks)
- (d) How would you interchange the **third** and **forth** lines in a document? (2 marks)
- (e) Explain the purpose of the **print preview** feature of your word-processing program. (2 marks)
- (f) State the line spacing used in the document. (1 mark)

Total 10 marks

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15. The following questions relate to Spreadsheets.

	A	B	C	D	E
1	PHONE MODEL	PRICE	IN STOCK	SOLD	REMAINING
2	MB607	14.99	4200	2200	2000
3	MB613	18.99	3650	1867	1783
4	MB614	18.99	4045	2245	1800
5	CP292	19.99	2634	1250	1384
6	MB612	110.99	3201	1664	1537
7	MB618	112.75	2876	1135	1741
8	CP307	116.99	1745	900	845
9	CP305	312.75	2255	200	2055
10					
11	Total	726.44	24606	11461	13145

The spreadsheet above indicates the number of cellphones sold during the period 2007.

- (a) How many rows and columns are in the spreadsheet? (2 marks)
- (b) Write the RANGE that shows the number of phones sold for EACH model. (1 mark)
- (c) State the **cell location** and **write the formula** that will compute the TOTAL number of cellphones remaining. (2 marks)
- (d) State which column of data should be formatted for currency. (1 mark)
- (e) If the value in cell C6 is changed, list TWO cells whose value will be affected, given that formulae are inserted to compute the totals. (2 marks)
- (f) What is the default **alignment** for:
 - (i) Numeric data? (1 mark)
 - (ii) Text data? (1 mark)

Total 10 marks

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16. The following questions relate to Database Management:

The owner of a small computer store maintains a stock card for EACH item in the store. A sample of the stock card is shown below:

Phone Model	Description	Quantity	Price	On Sale
MB607	ATT Flip Top	2209	14.99	No

- (a) Design the structure of a database table to correspond to the stock card. Your table structure should include appropriate Field Names, Field Type and Field Size. The data in the 'On Sale' field should only be Yes or No. (3 marks)
- (b) Which field in the table is the MOST appropriate to have a unique value? (1 mark)
- (c) State TWO ways in which the data in the table can be arranged. (2 marks)
- (d) List TWO meaningful queries that would be of benefit to the storeowner. (2 marks)
- (e) The storeowner needs a price list showing the Phone Model, Description and Price. Explain how this report can be generated. (2 marks)

Total 10 marks

END OF TEST

FORM TP 2010082



TEST CODE **01229020**

MAY/JUNE 2010

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours 15 minutes

13 MAY 2010 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. This paper consists of **THREE** sections and a total of **TWELVE** questions. Candidates **MUST** answer **ALL** questions in all **THREE** sections.
2. Number **EACH** answer correctly in the answer booklet.
3. Code is to be written in the programming language, **Pascal**.

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SECTION I

THEORY – 60 marks

Answer ALL questions in this section.

1. Figure 1 below shows the basic components of a computer system. Study Figure 1 and answer the questions that follow.

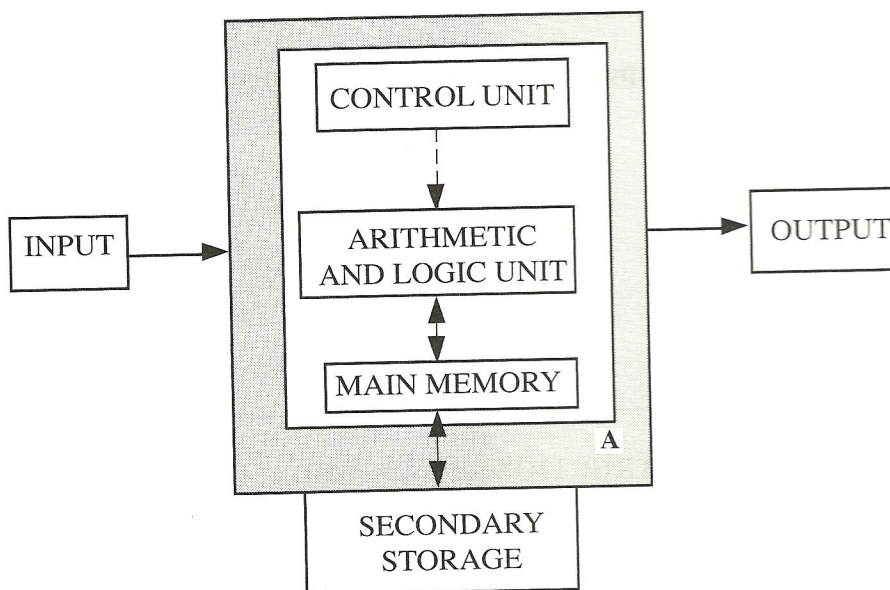


Figure 1

- (a) Name the component that performs EACH of the following functions:
- (i) The calculations and logic instructions (1 mark)
 - (ii) Stores programs and data that are currently being used by the CPU (1 mark)
 - (iii) Fetches instructions from memory, decodes them and processes them (1 mark)
 - (iv) Gets data into the shaded area A (1 mark)
 - (v) Sends processed data from the shaded area A (1 mark)
 - (vi) Keeps data and information for **future** use (1 mark)
- (b) Write the name of a component that
- (i) contains the Control Unit and the Arithmetic and Logic Unit (1 mark)
 - (ii) is also known as RAM (1 mark)
 - (iii) is also called a peripheral device. (1 mark)
- (c) Write the general name given to the collection of components in Figure 1. (1 mark)

Total 10 marks

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2. The following questions refer to Figures 2 and 3 below:

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\Administrator\dir_



Local Disk (C:)



DVD-RW Drive (E:)



Administrator's Documents



Shared Documents

Figure 2

Figure 3

- (a) State the type of user interface that is represented in EACH figure. (2 marks)
- (b) Indicate by writing the figure number of the interface that
- (i) uses a mouse
- (ii) would be easier for someone who is not familiar with a computer. (2 marks)
- (c) Give the name of another user interface. (1 mark)
- (d) State the name of the system software that provides the user interface. (1 mark)
- (e) You wish to change the name of a file you have on your computer. Explain how you would do so using the interface in
- (i) Figure 1
- (ii) Figure 2. (4 marks)

Total 10 marks

3. (a) Write the numbers 1 to 5 on separate lines in your answer booklet. Using the information in Table 1, match the number of the device with the letter for the task that the device performs by writing the letter next to the number in your answer booklet.

TABLE 1

	Device		Task
1	OMR	A	Reading magnetic ink characters on a check
2	Sensor	B	Reading codes with name and price information
3	MICR	C	Converting paper-based data into digital form
4	Scanner	D	Marking students' responses on a multiple-choice examination
5	Barcode reader	E	Turning on the fan when the environment is too hot

(5 marks)

- (b) For EACH of the devices listed in part (a), state ONE organization where each would MOST likely be used. (5 marks)

Total 10 marks

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4. After a storm, there was water damage to the files left on employees' desks, and power surges caused damage to computer hardware. When a check was done, the manager noticed that a number of computers were damaged and two hard drives, along with important files, were missing from computers in the accountant's office.
- (a) State ONE natural disaster that can cause damage to computer hardware, physical files or data. (1 mark)
 - (b) Using the scenario described above, for EACH of the following clearly indicate what type of theft or damage is present and provide TWO examples to justify your response:
 - (i) Deliberate theft OR deliberate damage to computers and files (3 marks)
 - (ii) Accidental theft OR accidental damage to computers and files (3 marks)
 - (c) Explain how EACH of the following should be protected from damage or theft:
 - (i) Physical files
 - (ii) Computer equipment
 - (iii) Software and data (3 marks)

Total 10 marks

5. E-learning, e-commerce and e-mail are very popular ways to access and share information.
- (a) State the word that the abbreviation 'e' represents in e-learning, e-commerce and e-mail. (1 mark)
 - (b) State the network of networks that is necessary to transfer this information to users across the world. (1 mark)
 - (c) State ONE advantage and ONE disadvantage of using any ONE of these methods (e-learning, e-commerce or e-mail) to access or share information. (2 marks)
 - (d) Write the appropriate term, e-learning, e-commerce or e-mail, that matches EACH of the following examples:
 - (i) Paying a fee to access the latest Top-40 tunes
 - (ii) Accessing a webpage to post a question for a tutor
 - (iii) Sending a single message to four friends (3 marks)
 - (e) Explain the difference between 'e-learning' and 'e-commerce'. (3 marks)

Total 10 marks

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6. The Power and Light Company has recently purchased a payroll information processing system.

- (a) State TWO advantages of an information processing system. (2 marks)
- (b) State TWO types of processing carried out by a payroll system. (2 marks)
- (c) State TWO types of reports generated by the payroll system. (2 marks)
- (d) The payroll system uses a master file and a transaction file. Explain what is contained in EACH file. (2 marks)
- (e) Records can be organized in a file in a serial or sequential manner. Explain the difference between the two methods of file organization. (2 marks)

Total 10 marks

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SECTION II

PRODUCTIVITY TOOLS – 15 marks

Answer ALL questions.

7. A beverage manufacturer uses a database that contains two tables to store information on customers.

TABLE 1: CATEGORY

Type	Name
D	Distributor
W	Wholesaler
R	Retailer

TABLE 2: CUSTOMER

CNO	CNAME	Type	Outlets	YTD SALES
D100	Bish and Sons	D	5	300,000.00
R100	Saywack	R	1	25,000.00
W100	Corbin	W	2	75,000.00
D200	Hamilton	D	3	100,000.00
W200	Singh	W	3	65,000.00
R200	Mohan	R	2	50,000.00
W300	Peter	W	2	30,000.00
W400	Pauline	W	1	20,000.00
R300	McDonald	R	2	30,000.00
D300	Narine	D	4	250,000.00

- (a) List TWO properties of a field. (2 marks)
- (b) Explain how you would delete a field from a table. (2 marks)
- (c) Give the name of the **primary** key in the CUSTOMER table. (1 mark)
- (d) Write a query to find ALL distributors with YTD SALES **greater** than 50,000.00. (3 marks)

Total 8 marks

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8. The accounts department of the beverage company has prepared a summary sheet showing beverage sales for 2009. This summary is given in the spreadsheet listed below:

	A	B	C	D
1	Type	Sales	Status	VAT
2	Soda	150,000.00		
3	Beer	175,000.00		
4	Malta	23,000.00		
5	Shandy	135,000.00		
6	Stout	75,000.00		
7	Total	558,000.00		

- (a) List TWO types of data that can be entered into a spreadsheet cell. (2 marks)
- (b) Write the formula that was inserted in cell B7 to compute the TOTAL sales. (1 mark)
- (c) What formatting feature was used on the Sales figures? (1 mark)
- (d) All beverages with sales under 100,000.00 are given a status of 'POOR' while others are given a status of 'GOOD'. Write the function that should be entered into cell C2 to show the status of the beverage 'soda'. (3 marks)

Total 7 marks

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SECTION III

PROBLEM SOLVING AND PROGRAMMING – 45 marks

Answer **ALL** questions.

9. (a) Table 3 shows labels in Column 1 and statements that represent the steps in problem solving in Column 2.

Arrange the labels to show the **correct** order in which the steps in problem solving occur. Write that order in your answer booklet.

TABLE 3

Labels	Statements
S1	Select the most efficient solution.
S2	Test and validate the solution.
S3	Definition of the problem.
S4	Develop the algorithm.
S5	Propose and evaluate solutions.

(5 marks)

- (b) You are using a program to calculate the average number of storms that have recently passed through the Caribbean. This is calculated by adding the number of storms and then dividing by the number of years. The screen for the program looks like the one below:

STORM STATS

Enter the Number of Years
(1 to 4)

Enter the Number of Storms:

Year 1	Year 2	Year 3	Year 4

Average number of storms:

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You were told that the number of storms for the last three years were 10, 8, and 12.

- (i) State the data that is required to input
 - a) the number of years
 - b) Year 1
 - c) Year 2
 - d) Year 3
 - e) Year 4. (5 marks)
- (ii) State the result for the 'Average number of storms'. (1 mark)
- (iii) Write an algorithm using pseudocode to input the total number of storms for three years, to calculate the average number of storms, and to output result. (4 marks)

Total 15 marks

10. (a) Each of the following programming statements contains ONE error. Rewrite the statement, correcting EACH error.

- (i) IF (X = 3) TRY Y :=8;
- (ii) WHILE (X = 3) UNTIL Y := Y + 1;
- (iii) FOR X :=1 – 10 DO
Writeln(X); (3 marks)
- (b) (i) Explain the difference between a 'variable' and a 'constant'. (2 marks)
- (ii) Write an example of a constant using a Pascal statement. (1 mark)
- (c) Write an example of a value that can be stored in EACH of the following types of variables:
 - (i) Integer
 - (ii) Real
 - (iii) Character
 - (iv) String (4 marks)

Total 10 marks

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11. (a) Rewrite the following fragment of code using the given control structures:

```
Mark :=1;  
While Mark <= 40 DO  
Begin  
    Writeln(Mark);  
    Mark := Mark + 1  
End;
```

- (i) a FOR loop (3 marks)
- (ii) A Repeat loop (4 marks)
- (b) Complete the following by replacing the letters A, B, C and D with the appropriate programming terms.

Rashida wrote a program in Pascal, and used a A to translate all program instructions at one time. The program produced a list of B which suggested that she made mistakes in the language rules of sentence structure when she was writing the program. Luckily, she did not have any C which meant that the sequence of programming statements was correct. Rashida took a long time D the code, since she could not find the errors in her source code. Eventually, her program was working. (4 marks)

- (c) (i) State TWO reasons why it is important to document a program. (2 marks)
- (ii) Explain the difference between 'system documentation' and 'user documentation'. (2 marks)

Total 15 marks

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12. Consider the following program in Pascal, and answer the questions that follow:

```
Program HUM (input, output);  
VAR CSEC:array[1..8] of integer;  
    Start, inc: integer;
```

```
Begin
```

```
    Inc := 0;
```

```
    FOR Start := 1 TO 8 DO
```

```
        CSEC[Start] := Inc;
```

```
End.
```

- (a) State the name of the program. (1 mark)
- (b) State the name of the array. (1 mark)
- (c) State the size of the array. (1 mark)
- (d) Explain the purpose of the array. (2 marks)

Total 5 marks

END OF TEST